

Designing machines
to see and move.
THAT'S SMART.

Building the vision
for fully integrated and
automated factories?
THAT'S INTELLIGENT.

ARROW
Five Years Out

The future is intelligent. Let Arrow guide you there.
See what's possible at [Arrow.com](https://www.arrow.com)



of transmitting and storing all raw data, systems become more efficient when they prioritise meaningful information, such as key metrics, threshold breaches, and aggregated summaries. This reduces long-term storage demands while preserving the data required for monitoring and decision-making.

Edge processing plays a major role in improving system efficiency. By processing data directly on the device or close to the source, organisations can reduce the need to transmit large volumes of raw data to the cloud. This lowers bandwidth usage, reduces dependence on cloud infrastructure, and improves overall system responsiveness. It also enables faster local decision-making without waiting for cloud-based computation.

On the operational side, connected products require strict governance over firmware updates, manufacturing changes, and feature roll-outs. Without controlled

approval and validation processes, unverified updates can propagate to devices in the field, leading to system failures. Structured release management ensures that changes are tested, validated, and safely deployed across all devices.

Field reliability depends heavily on how effectively failure feedback is handled. Issues observed in real-world environments must be systematically collected and analysed. A structured root-cause analysis process helps trace failures to their origins, enabling corrective actions that prevent recurrence in future releases.

Finally, platform design decisions have long-term implications for cost and flexibility. Heavy dependence on a single cloud or IoT platform can create vendor lock-in, increasing migration difficulty and costs over time. Designing portable architectures helps reduce this risk, allowing systems to shift between infrastructure providers when

required without major redesign efforts.

In hardware development, most costs don't show up early but appear later during manufacturing and real-world use. Small decisions in design, testing, component choice, and system architecture can lead to higher costs when the product is scaled or deployed. Manufacturing introduces challenges such as supply issues, production efficiency, and warranty support, while connected products also entail ongoing costs for cloud storage, updates, and maintenance. Overall, hardware cost control depends on making careful design decisions from the beginning so that the product can scale and run smoothly without unexpected expenses later. **EFY**

Co-author Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.